[Last updated: Oct 19th, 2026]

Thomas Hsiao

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Research Interests

Causal inference, spatiotemporal statistics, dimension reduction, informative sampling
Applications: Environmental health, electronic health records, disease surveillance

Education

Emory University, Atlanta, Georgia, USA
Ph.D. in Biostatistics — May 2025
Dissertation Advisor: Lance Waller
M.S. in Biostatistics — May 2024

Rice University, Houston, Texas, USA
B.A. in Statistics — May 2017
Capstone Advisor: Rudy Guerra

Professional Experience

Postdoctoral Researcher — June 2025 – Present
Department of Biostatistics and Bioinformatics
Emory University, Atlanta, Georgia, USA
Advisors: Howard Chang & Razieh Nabi

Post-Bachelor Fellow — Aug 2017 – Aug 2019
Institute for Health Metrics and Evaluation
University of Washington, Seattle, Washington, USA

Publications

Preprints and Manuscripts Under Review

Hsiao TW, Chang HH, & Nabi R (2026). Causal Sufficient Dimension Reduction for Multiple Continuous Exposures with an Application to Environmental Mixtures. arXiv: 2606.14840.

- An earlier version received the 2026 ENAR Distinguished Student Paper Award.

Li C, **Hsiao TW**, Ebelt S, Zhang R, Chang HH (2026). Temperature and Respiratory Emergency Department Visits: A Mediation Analysis with Ambient Ozone Exposure. arXiv: 2602.16970.

Pan S, **Hsiao TW**, Dunlop AL, Brennan PA, Chang HH, Everson T, Tan Y, Liang D, Eick SM (2026+). Uncovering chemical and non-chemical stressor interactions during pregnancy in relation to preschooler behavior problems among African American mother-child pairs.

Peer-Reviewed Publications

Li C, **Hsiao TW**, Warren JL, Darrow LA, Strickland MJ, Russell AG, Chang HH (2026). Maternal ambient air pollution and risk of stillbirth in Georgia, USA. *Environmental Epidemiology*, 10(4), e505.

Hsiao TW & Waller LA (2026). Inverse sampling intensity weighting for preferential sampling adjustment. *Journal of Agricultural, Biological, and Environmental Statistics*. Advance online publication.

Hsiao TW, Fede L, Gocke C, Waller LA, Ali MK, Varghese JS (2026). County Prevalence, Awareness, and Control of High Blood Pressure from Health Kiosks in the United States, 2017–2024. *Hypertension*, 83(2), e25829.

Hsiao TW, Gaskins AJ, Warren JL, Darrow LA, Strickland MJ, Russell AG, & Chang HH (2026). A time-to-event analysis of the association between ambient air pollution and risk of spontaneous abortion using vital records in the U.S. state of Georgia (2005-2014). *American Journal of Epidemiology*, 195(1), 60–69.

Varghese JS, **Hsiao TW**, Fede L, Chizak C, Gocke C, Waller LA, Ali MK (2026). National, state, and county estimates of adult overweight and obesity from electronic health records and kiosks in retail locations. *Obesity*, 34(4), 908–916.

Hsiao TW, Liao CY, Waller LA, Paynabar K (2025). Balancing access, precision, and equity in adaptive test site allocation with an application to COVID-19 in Atlanta, Georgia. *Scientific Reports*, 15, 32775.

Hsiao TW, Varghese JS, Fede L, Gocke C, Waller LA, & Ali MK (2025). Blood Pressure Measurements from Self-Service Health Kiosks in US Retail Stores, 2017–2024. *JAMA Cardiology*, 10(10), 1044–1048.

- **Highlighted in an Editor's Note**: Khan SS, Harrington RA, & Ahmad FS (2025). Blood Pressure Measurement in Retail Stores and Population Health. *JAMA Cardiology*, 10(10), 1049.

Kendrick PJ, *et al.* (2021). Spatial, temporal, and demographic patterns in prevalence of chewing tobacco use in 204 countries and territories, 1990-2019: a systematic analysis from the Global Burden of Disease Study 2019. *The Lancet Public Health*, 6(7), 482–499.

Reitsma MB, *et al.* (2021). Spatial, temporal, and demographic patterns in prevalence of smoking tobacco use and attributable disease burden in 204 countries and territories, 1990-2019: a systematic analysis from the Global Burden of Disease Study 2019. *The Lancet*, 397(10292), 2337–2360.

Vollset SE, Goren E, Yuan CW, Cao J, Smith AE, **Hsiao T**, *et al.* (2020). Fertility, mortality, migration, and population scenarios for 195 countries and territories from 2017 to 2100: a forecasting analysis for the Global Burden of Disease Study. *The Lancet*, 396(10258), 1285–1306.

Stanaway JD, *et al.* (2018). Global, regional, and national comparative risk assessment of 84 behavioural, environmental and occupational, and metabolic risks or clusters of risks for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet*, 392(10159), 1923–1994.

Murray CJL, *et al.* (2018). Population and fertility by age and sex for 195 countries and territories, 1950–2017: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet*, 392(10159), 1995–2051.

Lozano R, *et al.* (2018). Measuring progress from 1990 to 2017 and projecting attainment to 2030 of the health–related Sustainable Development Goals for 195 countries and territories: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet*, 392(10159), 2091–2138.

Roth GA, *et al.* (2018). Global, regional, and national age–sex–specific mortality for 282 causes of death in 195 countries and territories, 1980–2017: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet*, 392(10159), 1736–1788.

Anderson A, **Hsiao T**, & Metsis V (2017). Classification of Emotional Arousal During Multimedia Exposure. In *Proceedings of the 10th International Conference on PErvasive Technologies Related to Assistive Environments*, 181–184.

Wu JH, Simonette RA, **Hsiao T**, Doan HQ, Rady PL, & Tyring SK (2015). Cutaneous Human Polyomavirus Small T Antigens and 4E-BP1 Targeting. *Intervirology*, 58(6), 382–385.

Online Publications

Bill and Melinda Gates Foundation (2018). 2018 Goalkeepers Report. *The Goalkeepers Report*.

Hsiao T, DeLira R, Cruz P (2016). U.S.–Mexico Academic Mobility. *Baker Institute of Public Policy*.

Grants and Funding

Pending

NIEHS K99/R00 (Proposed 2027–2031)
High-dimensional causal mediation for environmental exposures and adverse pregnancy outcomes
Role: PI
Submitted Feb. 2026 (Competitive Not-Discussed); resubmission planned Nov. 2026

NIH R01 (Proposed 2027–2032)
MATCH: Menopause Transition and Cardiometabolic Health
Role: Co-I; PI: Shivani Patel
Submitted Oct. 2026

NIH R01 (Proposed 2027–2032)
Heterogeneity in Type 2 Diabetes and Risk of Alzheimer's Disease and Related Dementias
Role: Co-I; PI: Jithin Varghese
Submitted Oct. 2026

Completed

NSF Graduate Research Fellowship Program (NSF–GRFP) Aug 2021 – May 2025

Emory Laney Graduate School Fellowship Aug 2019 – May 2024

Software

`csdr` `R` package for causal sufficient dimension reduction for multiple continuous exposures

`ISIW` `R` package for inverse sampling intensity weighting for preferential sampling adjustment

`gallery` `R` package for generating test matrices for numerical experiments

Presentations

Conference Oral

ECCRI – Invited Talk Oct 2026
County Prevalence, Awareness, and Control of High Blood Pressure from Health Kiosks 2017-2024
Emory Clinical Cardiovascular Research Conference, Atlanta, Georgia, USA

ISEE – Invited Symposium Talk Aug 2026
Causal sufficient dimension reduction for studying environmental mixtures
Munich, Germany

ENAR – Oral Contributed Session Mar 2026
Causal sufficient dimension reduction for multiple continuous exposures
Indianapolis, Indiana, USA

JSM – Topic Contributed Paper Session Aug 2025
Inverse sampling intensity weighting for preferential sampling adjustment
Nashville, Tennessee, USA

ENAR – Oral Contributed Session Mar 2025
When does geostatistical design matter? Insights into the effect of preferential sampling on the MLE
New Orleans, Louisiana, USA

CAPASUS – Oral Presentation Sep 2023
A time-to-event analysis of maternal ambient air pollution and spontaneous abortion in the US state of Georgia, 2005-2014
Atlanta, Georgia, USA

Conference Poster

Winship Scientific Symposium Nov 2026
Causal sufficient dimension reduction for environmental mixtures
Atlanta, Georgia, USA

ENVR Workshop Oct 2026
Causal sufficient dimension reduction for environmental mixtures
Houston, Texas, USA

ISEE Aug 2026
Temperature and Respiratory Emergency Department Visits: an Ozone Mediation Analysis
Munich, Germany

ACIC – Tom Ten Have Poster Competition May 2026
Causal sufficient dimension reduction for multiple continuous exposures
Salt Lake City, Utah, USA

JSM – Opening Night Poster Session Aug 2024
Adjusting for varying sampling effort in species distribution modeling of Western monarchs
Portland, Oregon, USA

49th Union World Lung Health Conference Oct 2018
Risk reduction in former smokers: an analysis from the Global Burden of Disease 2017 Study
The Hague, Netherlands

Other Presentations

GECC Methods Meeting Nov 2026 (upcoming)
Causal sufficient dimension reduction for estimating and interpreting environmental mixtures
Gateway Exposome Coordinating Center, University of Southern California (online)

EMERGE Lab Meeting Jul 2026
Causal sufficient dimension reduction for estimating and interpreting environmental mixtures
Emory University Gangarosa Department of Environmental Health

ENVISION Lab Meeting Mar 2026
Causal sufficient dimension reduction for estimating and interpreting environmental mixtures
Emory University Gangarosa Department of Environmental Health

Dissertation Defense Mar 2025
Statistical methods for understanding and addressing preferential sampling in model-based geostatistics
Emory University Department of Biostatistics and Bioinformatics

Dissertation Proposal Mar 2023
Methods for fast spatial inference under preferential sampling
Emory University Department of Biostatistics and Bioinformatics

Annual Gates Family Planning Seminar Aug 2018
Estimation of age-specific fertility rates and comparison to World Population Projections
Bill and Melinda Gates Foundation

Bloomberg Tobacco Advisory Group Meeting Jul 2018
Estimation of smokeless tobacco burden in the Global Burden of Disease Study 2017
Institute for Health Metrics and Evaluation

Risk Factors External Review Week Mar 2018
Tobacco attributable burden in the Global Burden of Disease Study 2017
Institute for Health Metrics and Evaluation

Oshman Engineering Design Showcase May 2017
Crisis intervention: the reproducibility of psychological science
Rice University

Professional Activities and Service

Conference Activities

(PENDING) Organizer & Chair – JSM Invited Paper Session Aug 2027
Causal Inference for Continuous Exposures in the Real World

Chair – ISEE Invited Symposium Sept 2026
Contemporary Issues in Estimating Joint Health Effects of Multiple Exposures: New Applications and Methods

Chair – ENAR Oral-Contributed Session Mar 2026
Advancements in Spatial-Transcriptomics Analysis and Modeling Techniques

Member – ENAR Early Career Workshop Series (ECWS) Mar 2026

Peer Review

Journals: *American Journal of Epidemiology, BMC Public Health, Environment International, Environmental Research, Environmetrics, Epidemiology, International Journal of Health Geographics, PLOS One, Spatial Statistics*

Honors and Awards

2026 ENAR Distinguished Student Paper Award
2026 Emory Postdoctoral Academic Appreciation Award
2024 Livingston Award (Top PhD Student in Emory BIOS Department)
2023 Certificate of Excellence from 2023 CAPASUS Conference
2023 2nd Place Emory BIOS Senior Student Presentation
2019 Emory Laney Graduate School Fellowship
2016 NSF Research Experience for Undergraduates (REU) Award
2015 Molecular Basis in Infectious Disease (MBID) Training Grant
2015 UTSA Honors College Dean's Scholarship
2012 Fountainhead Essay Contest National Semifinalist

Teaching Experience

Teaching Assistant – Spatial Statistics in Epidemiology and Public Health
SISMID · Emory University 2025
SISMID · University of Washington 2023

Instructor – BIOS Theory Qualifying Exam Review 2022–2024
2nd Year Theory Qualifying Exam · Emory University

Tutor – Second-Year Biostatistics Coursework 2023
Emory University

Teaching Assistant – Advanced Statistical Inference 2022
BIOS 711 · Emory University

Teaching Assistant – Advanced Linear Models 2021
BIOS 707 · Emory University

Teaching Assistant – Probability Theory 2021
BIOS 512 · Emory University

Teaching Assistant – Real Analysis and Linear Algebra 2021
Incoming PhD Bootcamp · Emory University

Teaching Assistant – R Programming 2020
BIOS 541 · Emory University

Instructor – Applied Modeling 2018
Bootcamp · Institute for Health Metrics and Evaluation

Grader – Computational Finance I: Market Models 2016
STAT 486/686, Rice University

Supplemental Instructor – General Chemistry I 2014
CHE 1103, University of Texas at San Antonio